

Jumping Into the Frame

Reconstructing Interactive, Physically-Grounded 3D Environments from a Single Image

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A photograph is not a place

MOTIVATION

- Recovering 3D from one view is underconstrained: infinitely many scenes project to the same image.
- People read a photograph as a place anyway — the ground continues past the frame, the occluded side of a rock exists, the flowers are individual plants.
- Photographs are among the most abundant records of place we have.
- The goal: an environment you can *jump into*, not a picture you can orbit.



Four problems that compound

THE CHALLENGE

- **Extreme ill-posedness.** Roughly 5/6 of the sphere is never observed, and everything behind every foreground object is unknown.
- **Compounding error.** Depth, then geometry, then materials, then objects. A discrepancy invisible at one stage is structural by the next.
- **Heterogeneous content.** A height field suits a mountainside, a mesh a tree, a scattered population a meadow, an animated surface the sea — and they must share one coordinate frame, scale and lighting.
- **Interactivity budget.** Stereo rendering at headset frame rates constrains every choice, hardest on the animated elements that make it interactive.

Prior work stops at viewable

RELATED WORK

- Single-image scene generation has advanced fast: CubeDiff, DreamCube, PanoDreamer, LayerPano3D, WorldGen, SphericalDreamer.
- These optimize visual fidelity from viewpoints near the original camera.
- The output is a panorama, a layered mesh, or a radiance field — something to look at.

What none of them return: a collidable surface, a population whose density can be sampled, a water plane separable from its bed, or a mesh with a skeleton in it.

One idea: reconstruct by physical role

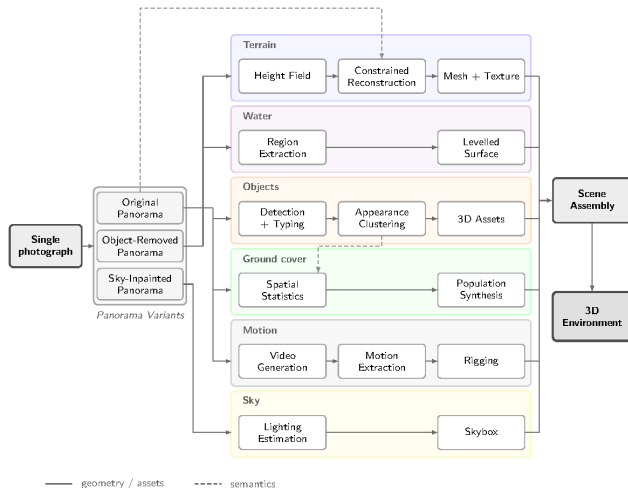
APPROACH

A scene is not one thing. Ground, discrete objects, ground cover, water and sky differ in how they must be reconstructed, represented, textured, and how they behave at runtime.

- **Terrain** — a continuous height field carrying collision
- **Objects** — meshes reconstructed once per appearance group, then instanced
- **Ground cover** — a scattered population, too numerous to place individually
- **Water** — a separate leveled surface with an animated shader
- **Motion** — rigid bodies or anchored sway, per object
- **Sky** — skybox and image-based lighting source

The pipeline

42 STAGES, 38 ACTIVE



One photograph becomes a sphere

THE PANORAMIC SUBSTRATE



Input: about a sixth of the sphere

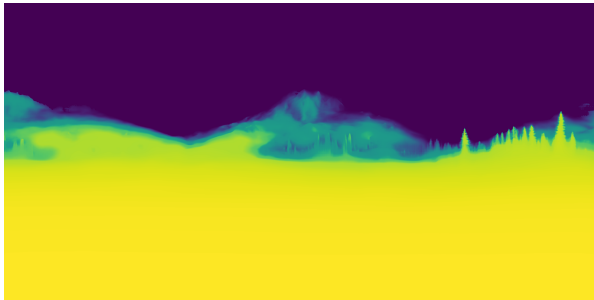


Generated panorama: the other ~83% is synthesized

- WorldGen by default; FLUX, CubeDiff and DreamCube are interchangeable backends.
- The panorama is an intermediate representation, not the deliverable.

Depth is relative — calibration makes it metric

THE PANORAMIC SUBSTRATE



- Panoramic depth is relative. We fit a monotonic curve against Depth Anything 3 on the original photo, over the region it observed.
- 14 quantile bins, median knots, running-maximum monotonicity.
- Beyond the fitted range we extrapolate in log-depth, capped at $20\times$.

Angularly faithful but metrically compressed. Near distances are accurate, far ones systematically shrunk — and object size, computed as angular extent times depth, inherits that shrinkage eight stages later.

Two panoramas, because they answer different questions

THE PANORAMIC SUBSTRATE



Original — what the scene contains

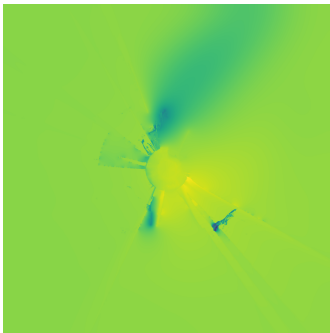


Object-removed — what the ground looks like

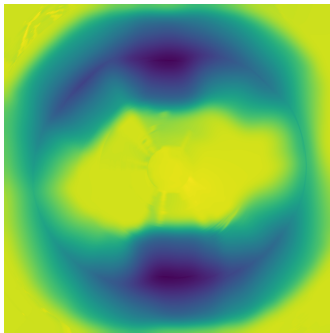
Depth from the original would place the flower canopy, not the ground, at the bottom of the height field. Geometry reads the object-removed panorama; semantics reads the original.

Completing terrain that was never seen

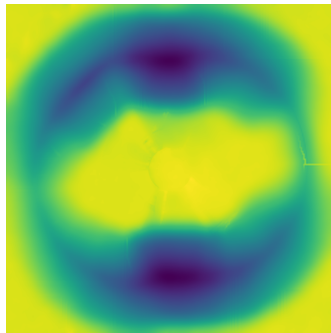
TERRAIN GEOMETRY



Projected observations — 7% of
the grid



Harmonic completion



Fluvial erosion

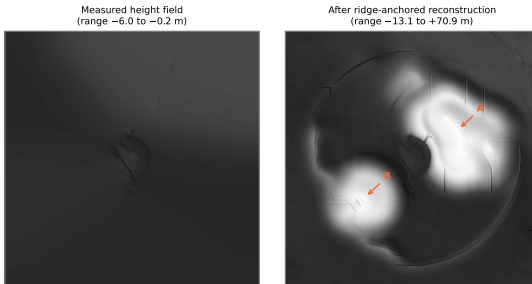
Measured cells become Dirichlet boundary conditions and are never modified. The unknown region is the smoothest surface consistent with them — then Landlab's fluvial incision gives it drainage structure, so invented relief is shaped by the process that shapes real landscapes.

The sky boundary is assumed to be ground

TERRAIN — THE KEY ASSUMPTION

- Past a few hundred meters the depth model has saturated. The silhouette is the only signal left.
- For each panorama column, the first non-sky pixel gives an elevation angle, placed at a fixed radius.
- This works whenever the horizon really is a landform.

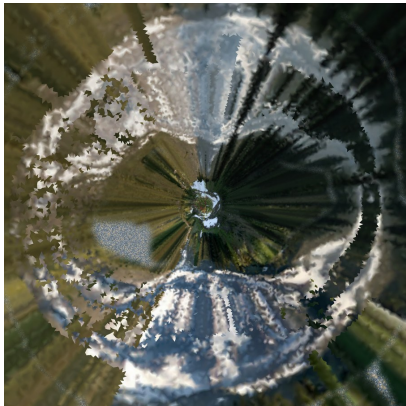
In Paris the horizon is a roofline 40 m away. A flat height field (-6.0 m to -0.2 m) became -13.1 m to 70.9 m — a hill on the cathedral.



Paris height field: measured (left) vs after ridge-anchored reconstruction (right)

Projecting the photo onto the ground fails

TERRAIN APPEARANCE



- **Grazing incidence.** At 80 m one panorama row covers roughly 12 m of ground. The result is radial smearing.
- **Upright content.** A 0.5 m flower spans a range of elevations, each landing at a different radius — one flower becomes a streak meters long.
- **The fix.** A layered splat material: real photographic color where projection is trustworthy, pattern-textured real material where it is not, generated tiles only as a last resort.

Detect, cluster, reconstruct once, instance everywhere

OBJECT DECOMPOSITION



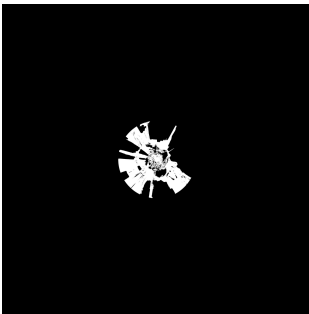
- SAM 2 segments the full 360°; CLIP classifies; Grounding DINO verifies by having to *find* the label in the crop.
- DINOv2 sub-clusters each class into appearance variants. One mesh per (class, bucket) group, instanced at every member.
- **Three representations:** reconstructed mesh → crossed card (three fixed planes) → camera-facing billboard, chosen on projected angular size.

A detector finds tens of flowers in a meadow of thousands

POPULATION SYNTHESIS



Synthesized tree population

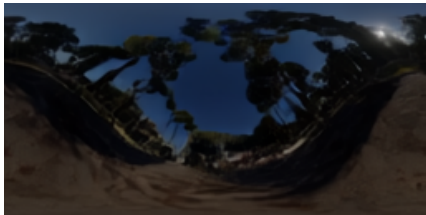


Ground-cover mask

- **Synthesis** matches the pair-correlation statistics of real detections — a point process, not a fixed density.
- **Ground cover** has no discrete instances to generalize from. It is scattered over a mask, vetoed by excess-green so snowfields do not become meadow.

Estimate the light, then remove it from every asset

LIGHTING

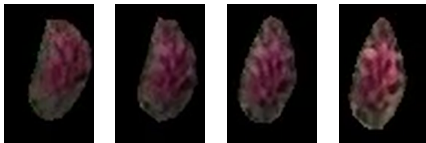


Estimated HDR environment (log scale)

- LuxDiT recovers an HDR environment and a dominant light direction.
- Every asset is cut from a photograph, so it already carries the light of the moment it was taken.
- IntrinsicDiffusion separates albedo from shading, so the client relights material color rather than a photograph.
- This is what makes assets portable: an instance rarely sits where it was photographed.

Motion from a generated video

ANIMATION

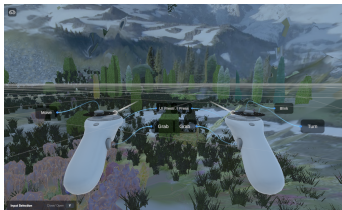


One tracked flower: 0 s, 1.2 s, 2.3 s, 3.9 s

- LTX-2 generates a short video with the camera locked off, so centroid drift is the object moving.
- Drift under 60% of its own bounding-box diagonal means anchored: it gets a sway amplitude and frequency, and a three-bone rig.
- Anything drifting further is a rigid body — velocity and acceleration handed to the physics engine.
- One scene-global wind field, so a gust crosses the meadow and everything in its path responds in turn.

Three captures, chosen to break different assumptions

RESULTS



Mount Rainier — alpine meadow



Paris — flat urban river scene



Shark Fin Cove — coastal cliffs

No ground-truth 3D exists for any of them, so evaluation is diagnostic: instrument the pipeline, measure intermediates against what the scene demonstrably contains, localize defects to stages.

Every failure localized to an assumption, not a model

RESULTS

Skyline read as landform	Paris: 0% of horizon columns type terrain, against 98% and 100% elsewhere
Confident misclassification	Shark Fin: 49 detections at median confidence 0.95, almost all rock texture
Thin vegetation	SAM 3D returns aspect 3.50:1 against detections at 0.12:1 — a $28\times$ disagreement
Water follows terrain	$11\,872\text{ m}^2$ of sea spanning 73.5 m in elevation — the ocean draped up the cliffs
Far-field texture	67.8% of Shark Fin's terrain exceeds 0.5 m per texel, against 20.9% for Rainier

Each was fixed by making its premise testable from evidence the pipeline already computes — not by tuning a threshold or swapping a model, and measured on every capture rather than only the one that broke.

The transferable result is methodological

WHAT WE LEARNED

Into Frame mixes learned models and classical graphics roughly two-to-one by stage count. Neither part would produce this output alone.

GENERATIVE MODELS

Make the problem tractable at all. Five sixths of the sphere and the ground behind every object are never observed, and no classical reasoning recovers content absent from the input.

They answer *what is there*.

Of 38 active stages, 12 use no learned model at all — and those are disproportionately the ones that produce the scene's physical behavior.

CLASSICAL GRAPHICS

Supply the structure. A harmonic solve that never modifies measured ground, fluvial incision, point-process synthesis, pattern texturing, a rig driven by a shared wind field.

They decide *how it behaves*.

Where this leaves us

CONCLUSIONS & FUTURE WORK

- A single photograph becomes an interactive environment with terrain, objects, ground cover, water and lighting as distinct physical components.
- The decomposition buys the depth of the result — each component behaves according to its role.
- Remaining failures are assumptions that held for outdoor natural landscapes and did not extrapolate.

The boundaries are legible and measurable — addressing them means making implicit commitments explicit, rather than waiting for better models.

FUTURE WORK

- Explicit scene classification, consumed by every stage embedding an implicit assumption
- Confidence propagation, so the system can report where it is guessing
- Species-aware reconstruction for thin structures
- Joint refinement across depth, geometry and placement
- Perceptual evaluation in the headset
- Spatialized audio and richer interaction